NAVDEEP SINGH JAKHAR

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EDUCATION

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY, KATTANKULATHUR - CHENNAI, TN — **May 2025**

- B.Tech in CSE with specialization in IoT, **CGPA: 9.59**
- **Awards and Honors:** Performance-based scholarship; **Dept. Rank - 5**
- **Extra-curriculars:** Editorial Domain Founder and Lead in SRM Team Humanoid, Editorial Lead in SRMKZILLA

RESEARCH EXPERIENCE

SPACE APPLICATIONS CENTRE - INDIAN SPACE RESEARCH ORGANISATION (ISRO), GJ — **Oct 2024 - June 2025**

Deep Learning Researcher (Research Focus: Learned Image Compression)

Supervisor: *Jimit J Gadhia (Scientist/Engineer - SG)*

Project-1: Learned Image Compression for Satellite Systems

- Reimplemented foundational learned compression models (Ballé et al. 2017, 2018) using tensorflow_compression, focusing on end-to-end entropy bottleneck modeling and rate-distortion optimization.
- Built a fully modular training and evaluation pipeline with CLI support, supporting both standard and custom datasets.
- Benchmarked multiple architectures (Ballé hyperprior, MBT, Cheng2020) against classical codecs (JPEG 2000), producing comparative rate–distortion curves.
- Initiated domain adaptation for learned compression; fine-tuned VAEs and planned extensions to satellite imagery.
- Delivered a technical lecture to ISRO scientists titled “Learned Image Compression: Basics to State-of-the-Art”, covering theoretical foundations, advances in the field, and implementation strategies.

Project-2: Multi-Class Satellite Terrain Classification under Imbalanced Data

- Built a satellite image classifier using ResNet50V2, trained on 256×256 patches from 6400×6400 images.
- Addressed class imbalance using a custom weighted categorical cross-entropy loss, leading to 9% increase in accuracy.
- Applied early stopping, learning rate scheduling, and checkpointing for stable model convergence.
- Quantized the model post-training, maintaining accuracy while optimizing for edge inference on Xilinx ZCU104.

CARNEGIE MELLON UNIVERSITY, Pittsburgh, PA, USA — **June 2024 - July 2024**

AI Research Intern

Supervisors:

Dr. Raj Reddy (Moza Bint Nasser University Professor, School of Computer Science, Carnegie Mellon University, PA)
Dr. Naveena Yanamala (Assoc. Professor of Medicine, Director of DS/ML Research, Rutgers RWJ Medical School, NJ)

Project: Cardiac Event Prediction

- Created a predictive multi-modal framework for aortic stenosis risk, integrating diverse data sources, which resulted in an 8% accuracy increase compared to previous work.
- Used deep learning to improve predictive accuracy in cardiac event forecasting through extensive feature engineering.
- Utilized microcontrollers and hardware for practical implementations and data collection.

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY, Chennai, TN

Undergraduate Researcher

Supervisor: *Dr. M. Thenmozhi (Professor, SRM Institute of Science and Technology, KTR, Chennai - TN)*

Project-1: Public Transport Network Optimization using Digital Twin — **June 2024 - Jan 2025**

- Designed a framework to optimize public transport stop placement and route design in metropolitan areas.
- Estimated reduction in average travel time of almost 30% in congested metropolitan areas.
- Increased 5-minute walk accessibility to bus stops from 72% to 89% of the population.

Project-2: Multimodal Diagnostic Framework for Autism Spectrum Disorder (ASD) — **August 2024 - Jan 2025**

- Designed a multimodal diagnostic framework for Autism Spectrum Disorder (ASD) using machine learning and behavioral analysis to aid in accurate diagnosis and personalized care.
- Created classification models using eye-tracking data and facial images to predict ASD presence.

PROJECTS

- ***Reimplementations of Papers:***
    - **Ballé et al. 2018 "Variational Image Compression with a Scale Hyperprior"** [code] - Faithfully reimplemented the full model using TensorFlow and tensorflow_compression.
    - **Ballé et al. 2017 "End-to-End Optimized Image Compression"** [code] - Developed the full pipeline for entropy-coded learned compression using a factorized prior.
    - **Vision Transformer paper - Dosovitskiy et al. 2021** [code] - Reimplemented the Vision Transformer model, including the self-attention mechanism and position embeddings
- ***Cross-Domain Neural Models for Learned Compression (Ongoing)*** [code]**:**
    - Designed and implemented a Domain-Adaptive Residual VAE, trained on MNIST and fine-tuned on EMNIST, to study robustness under low-to-moderate domain shift.
    - Investigating cross-domain generalization using techniques like fine-tuning, domain-specific batch normalization, latent space alignment (adversarial loss or MMD), and entropy model calibration.
    - Currently extending domain adaptation analysis to more complex compression models, including Ballé et al. 2018 (Scale Hyperprior) and Ballé 2018 + Transformer Hyperprior.
- **Balle2018 + Transformer Hyperprior *(Ongoing)*** [code] **-** Extending the Ballé et al. 2018 compression model by replacing the traditional hyperprior with a Transformer-based entropy model.
- ***ETA and Route Prediction using Road Segment Analysis* -** Built a dynamic ETA and route prediction model for Bangalore using road segment analysis and OpenStreetMap data, with real-time updates based on GPS-enabled smartphone data.
- ***Cineflix*** [code] **-** Built a content-based movie recommendation system by scraping and analyzing Wikipedia film data to generate personalized suggestions.

WORK EXPERIENCE

NEWTON SCHOOL, Bengaluru, KA **Jan 2024 - April 2024**

Data Science Intern

- Developed advanced ML and DL models for analysis & educational purposes, contributing to curriculum enhancement.
- Conducted interviews as part of the admissions team to assess candidates' suitability for the program; led TA sessions.

SKILLS & PROFICIENCIES

- ***Specializations****:* Learned Image Compression, Probabilistic Models (VAEs, entropy models, etc.), Encoder-Decoder Architectures, Computer Vision, Natural Language Processing, Deep Learning
- ***Languages, Libraries, and APIs****:* C/C++, Python, TensorFlow, TensorFlow Compression, PyTorch (Basic), Scikit-learn, SQL
- ***Tools & Platforms****:* Docker, Git, Linux
- ***Languages****: Hindi* - Native proficiency, *English - Fluent (IELTS - 8.0; R-8.5, L-8.5, W-7.0, S-8.0)*

CERTIFICATIONS

- DeepLearning.AI TensorFlow Developer Professional Certificate
- AWS Academy Graduate - AWS Academy Machine Learning Foundations - AWS
- TensorFlow Advanced Techniques Specialization
- Mathematics for Machine Learning and Data Science Specialization - DeepLearning.AI (Ongoing)

PUBLICATIONS

1. Arnav Aggarwal, Ronit Kumar, **Navdeep Singh Jakhar**, Gouthaman P. (2023), "Live Tracking of Metro Rail Using Internet of Things," IEEE, 2023 International Conference on ISRACET
2. **Navdeep Singh Jakhar**, Ronit Kumar, Dr. Thenmozhi M (2025), “Optimising Public Transport Networks in Expanding Metropolitan Areas: A Framework for Stop Placement and Route Revamping”, Conference on Advances in Communication Networks & Systems (CoaCoNS 2025) - AIP Conference Proceedings **[ACCEPTED FOR PUBLICATION]**
3. **Navdeep Singh Jakhar**, Adeline Rajaselvam, K. Bhuvaneswari, and M. Thenmozhi (2025), “Domain-Adaptive Residual Variational Autoencoder for Efficient Image Reconstruction and Compression” [**UNDER REVIEW**]